9 Integrály pohybu

9.1 "Diferenciál" operátoru

$$\hat{A},\ \langle \hat{A} \rangle_\psi = \int_{\mathbb{R}} \bar{\psi}(r,t)\, \hat{A}(x,t)\, \psi(r,t)\, dx$$

[illegible]

$$\frac{d}{dt} \langle \hat{A} \rangle_\psi = \int_{\mathbb{R}} \left[\frac{\partial}{\partial t}\bar{\psi}\hat{A}\psi + \bar{\psi}\frac{\partial}{\partial t}\hat{A}\psi + \bar{\psi}\hat{A}\frac{\partial}{\partial t}\psi \right] dx$$

$\downarrow$ z ČSR $i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi$ | $\frac{\partial \psi}{\partial t} = \frac{1}{i\hbar}\hat{H}\psi$ | $\frac{\partial \bar{\psi}}{\partial t} = -\frac{1}{i\hbar}\hat{H}\bar{\psi}$

$$= \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi - \frac{1}{i\hbar} \langle \hat{H}\bar{\psi}, \hat{A}\psi \rangle + \frac{1}{i\hbar} \langle \bar{\psi}, \hat{A}\hat{H}\psi \rangle$$

samosdružený

$$= \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi + \frac{1}{i\hbar} \left[\langle \bar{\psi}, \hat{A}\hat{H}\psi \rangle - \langle \bar{\psi}, \hat{H}\hat{A}\psi \rangle \right]$$

$$= \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle_\psi + \frac{1}{i\hbar} \left\langle [\hat{A}, \hat{H}] \right\rangle_\psi$$

$\Rightarrow$ dává smysl zavést $\quad \frac{d}{dt}\hat{A} = \frac{\partial \hat{A}}{\partial t} + \frac{1}{i\hbar}[\hat{A}, \hat{H}]$

$$\frac{\partial \hat{A}}{\partial t} = 0 \Rightarrow \frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[\hat{A}, \hat{H}]$$

$\rightarrow$ souvisí s přesouváním derivací

$$\frac{d(\hat{A}+\hat{B})}{dt} = \frac{d\hat{A}}{dt}\hat{B} + \hat{A}\frac{d\hat{B}}{dt}$$

$\Rightarrow \quad \hat{x}, \hat{p}: \quad \frac{d\hat{x}}{dt} = \frac{1}{i\hbar}[\hat{x}, \hat{H}]$

$$\frac{d\hat{p}}{dt} = \frac{1}{i\hbar}[\hat{p}, \hat{H}]$$

$$\hat{H} = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + V(x,y,z,t)$$

$$\Rightarrow \frac{d\hat{x}}{dt} = \frac{1}{2mi\hbar}\left[\hat{x}\hat{p}_x^2 - \hat{p}_x^2\hat{x}\right] =$$

$$= \frac{1}{2mi\hbar}\left[(i\hbar + \hat{p}_x\hat{x})\hat{p}_x - \hat{p}_x^2\hat{x}\right]$$

$$= \frac{1}{2mi\hbar} 2i\hbar\, \hat{p}_x = \frac{\hat{p}_x}{m} \quad \leftrightarrow \quad \frac{d\langle \hat{x} \rangle_\psi}{dt} = \frac{\langle \hat{p}_x \rangle_\psi}{m}$$

$$\Rightarrow \frac{d\hat{p}_x}{dt} = \frac{1}{i\hbar}\left(\hat{p}_x\hat{V} - \hat{V}\hat{p}_x\right) = \frac{1}{i\hbar}\left(-i\hbar\frac{\partial \hat{V}}{\partial x}\right)$$

$$= -\frac{\partial \hat{V}}{\partial x} =: F_x \quad \leftrightarrow \quad \frac{d\langle \hat{p}_x \rangle_\psi}{dt} = -\left\langle \frac{\partial V}{\partial x} \right\rangle_\psi$$

$$\frac{d^2\langle \hat{x} \rangle_\psi}{dt^2} = \frac{d}{dt}\frac{\langle \hat{p}_x \rangle_\psi}{m} = \frac{1}{m}\langle \hat{F}_x \rangle_\psi$$

$$\Rightarrow \quad m\frac{d^2}{dt^2}\langle \hat{x} \rangle_\psi = \langle \hat{F}_x \rangle_\psi$$

9.3 Zákony zachování

$\hat{A}$ je integrálem pohybu $\quad \frac{d\hat{A}}{dt} = 0$

$$\frac{\partial \hat{A}}{\partial t} = 0 \Rightarrow [\hat{A}, \hat{H}] = 0$$

a) Volná částice – je ideál, který není dosažitelný

$$\hat{H} = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m}$$

$\Rightarrow \quad \hat{p}_x, \hat{p}_y, \hat{p}_z, \hat{H}$ jsou IP

$\rightarrow$ ZZH a ZZE

b) $\frac{d\hat{H}}{dt} = \frac{\partial \hat{H}}{\partial t} + \frac{1}{i\hbar}[\hat{H}, \hat{H}] = \frac{\partial \hat{H}}{\partial t}$

$\Rightarrow$ pokud $\hat{H}$ nezávisí na čase, pak je ZZE

c) Centrální pole

$$V(x,y,z,t) = V(r)$$

$\hat{H} = -\frac{\hbar^2}{2m}\Delta + V(r) \quad \Rightarrow$ přechod do sférických souřadnic

$x = r\sin\theta\cos\varphi \qquad r \in [0, +\infty)$

$y = r\sin\theta\sin\varphi \qquad \theta \in [0, \pi]$

$z = r\cos\theta \qquad \varphi \in [0, 2\pi)$

$$dx\,dy\,dz = r^2\sin\theta\, dr\, d\theta\, d\varphi$$

$$\Delta_{r,\theta,\varphi} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\Delta_{\theta,\varphi}$$

$$\Delta_{\theta,\varphi} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$

operátor hybnosti: $\quad \hat{L} = \hat{r} \times \hat{p} = (\hat{L}_x, \hat{L}_y, \hat{L}_z)$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

když vyjádříme $\hat{p}$ ve sfér. souř., vyjde:

$$\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y = i\hbar\left(\sin\varphi\frac{\partial}{\partial\theta} + \cot\theta\cos\varphi\frac{\partial}{\partial\varphi}\right)$$

$$\hat{L}_y = \hat{z}\hat{p}_x - \hat{x}\hat{p}_z = -i\hbar\left(\cos\varphi\frac{\partial}{\partial\theta} - \cot\theta\sin\varphi\frac{\partial}{\partial\varphi}\right)$$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = -i\hbar\frac{\partial}{\partial\varphi}$$

$$\Rightarrow \hat{L}^2 = -\hbar^2\Delta_{\theta,\varphi}$$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2mr^2} + V(r)$$

zjevně $\hat{L}^2$ komutuje s $\hat{H}$, jelikož nezávisí na r

$$\Rightarrow [\hat{L}^2, \hat{H}] = [\hat{L}_x, \hat{H}] = [\hat{L}_y, \hat{H}] = [\hat{L}_z, \hat{H}] = 0$$

$\rightarrow$ ZZMH $\Rightarrow$ Keplerovy zákony